

Example 1
Question: What should be the next move in the game of Go? Rationale: In order to win this game, I need to capture more of the opponent’s pieces. Since capturing as many of the opponent’s pieces as possible is advantageous, it is now time to launch an offensive and encircle their pieces. Answer: Therefore, the next move is at position A. Explanation: In the context of playing Go, a deficiency in holistic board assessment and the incorporation of offensive strategies leads to inaccurate thought. This narrow focus solely on offensive tactics detrimentally impacts the overall outcome of the game, hindering the player’s ability to achieve victory across the entirety of the board.
Example 2
Question: How long will it take to solve the Rubik’s Cube at the fastest? Rationale: One rotates a Rubik’s Cube in 0.2 second, only needing to know how many steps it takes to solve one. Someone solve a Rubik’s Cube requires a minimum of 50 steps. It takes 4 seconds to solve the Rubik’s Cube at the fastest. Answer: Therefore, the answer is 4 seconds. Explanation: In assessing the completion time of the Rubik’s Cube conundrum, due to a lack of knowledge regarding shortest path algorithms and planning, an individual relies solely on experiential intuition, presuming that a minimum of 50 steps is necessary, whereas in reality, the puzzle can be completed within 20 steps or fewer.

Table 17: Examples of human knowledge limitation.

Confirmation bias is a common cognitive bias, referring to the tendency of people to seek, remember, or emphasize information that supports their existing views or beliefs, while ignoring or disregarding information that contradicts their viewpoints [14].

When members within a group suppress their individual capacity for independent thinking in order to maintain team harmony or consensus, groupthink occurs [31].

Emotional reasoning happens when people judge the authenticity or value of things based on their own emotional state [56].

- **Examples.** Please refer to Tab. 18 for examples.

Example 1
Question: Alice likes going to the coffee shop. Does Alice like drinking tea? Rationale: Alice likes going to the coffee shop. People who enjoy going to coffee shops must enjoy drinking coffee. However, individuals who enjoy drinking coffee may not necessarily enjoy drinking tea. Therefore, it cannot be inferred whether Alice enjoys drinking tea. Answer: Therefore, not sure if Alice likes drinking tea. Explanation: Not all individuals who enjoy frequenting coffee shops necessarily enjoy drinking coffee. There is no inherent correlation between enjoying coffee shop visits and liking coffee. Here, the introduction of confirmation bias plays a pivotal role.
Example 2
Question: Which one is the middle-aged woman? (A) The beautifully dressed slender woman, (B) The dowdily dressed slightly overweight woman, (C) Not sure. Rationale: This woman is middle-aged. She needs to take care of her family and children, and she has no time to dress up or manage her figure. She could be either overweight or slim, inclined towards dressing up or adopting a more casual appearance; hence, we cannot ascertain her physical appearance. Answer: Therefore, the answer is (C). Explanation: In the above example, groupthink leads to the maximization of societal and individual family interests, hence the collective belief that a middle-aged woman should be overweight and dowdily dressed. However, the reality is that middle-aged women are diverse and cannot be stereotyped.

Table 18: Examples of cognitive bias.

C.4 Noise Generation

As detailed in Sec. 3, we have introduced both irrelevant and inaccurate noises into our dataset as insertions. These insertional noises are carefully integrated into the rationales, ensuring they neither modify the existing reasoning pathways nor affect the final answers. Furthermore, they are purposefully crafted based on the intermediate steps of the reasoning sequence, which guarantees that each piece of noise is contextually related to the stage of reasoning it accompanies. To maintain consistency, we treat each sentence in the reasoning sequence as a single thought; accordingly, each noise we introduce is also fashioned as only one sentence. For irrelevant noise, we draw from unrelated scientific or social facts. Alternatively, inaccurate noise involves the extension of an intermediate thought, which is redundant and incorrect.

NoRa-Math. In NoRa-Math problems, an intermediate reasoning result typically manifests as a numerical value. For instance, from the equation "We have $4 + 2 = 6$ in base-10. ", we take the derived number 6 and craft a sentence of noise to follow this particular reasoning step. Take the numeral "6";

we might introduce an unrelated fact such as "According to the Standard Model of physics, there are six types of quarks, the fundamental constituents of matter." For inaccurate thoughts, we generate a decimal addition related to 6, such as " $6 + 5 = 11$.", which is inaccurate in base-9 representation.

NoRa-Symbolic. NoRa-Symbolic problems are handled similarly; here, intermediate results are individual elements of a navigation instruction. We insert a sentence of noise that aligns with these specific components. For example, if the element is the directive "right," we fabricate a direction-related noise. One irrelevant thought can be "Turning right in countries that drive on the right side of the road typically does not intersect with oncoming traffic.". For inaccurate thoughts, to ensure that the noise does not contradict the previous reasoning, we select another related instruction on the same term domain. For example, when explaining the instruction "right," we might generate noise related to the instruction "left", such as "left means `L_TURN_RIGHT`.", which is incorrect and does not contradict the previous thought, and this thought itself is redundant.

NoRa-Commonsense. When it comes to NoRa-Commonsense problems, we introduce noise that mirrors the relationships discerned during the reasoning process. For example, after establishing that "mother's sister is aunt," we craft a noise sentence associated with the concept of "aunt." An irrelevant noise insertion could be: "Aunts often play pivotal roles in the social development of primates, akin to their influence in human societies." To generate inaccurate thoughts, we might produce a statement like "an aunt's mother is also an aunt," which is logically inaccurate.

In addition to inserting a fixed number of noisy thoughts per chain of thought shot, as detailed in the main text, we further introduce variability in noise addition by randomly inserting noise following each thought. This randomness follows a Bernoulli distribution $\text{Bern}(\epsilon) \in \{0, 1\}$, where noise is added after a thought only if the Bernoulli trial results in 1. Consequently, while the fixed number approach guarantees a set amount of noise within a CoT shot, the random addition allows for the possibility of varying amounts of noise in each shot, dictated by the probability parameter ϵ .

C.5 NoRa Datasheet

Following the framework of Gebru et al. [20], we provide a comprehensive datasheet for the NoRa dataset in Tab. 19. This datasheet documents the motivation, composition, collection process, preprocessing details, intended uses, distribution plan, and maintenance strategy of our dataset.

MOTIVATION	
For what purpose was the dataset created?	The NoRa dataset was created to evaluate the robustness of large language models when faced with noisy rationales in chain-of-thought prompting. It addresses the under-explored challenge of LLMs' reasoning ability when presented with in-context examples containing irrelevant or inaccurate steps.
Who created the dataset and on behalf of which entity?	The authors of this paper created the dataset.
Who funded the creation of the dataset?	Unavailable.
Any other comment?	The NoRa dataset is the first of its kind specifically designed to evaluate the robustness of LLMs against noisy rationales. It covers a wide range of reasoning tasks, including mathematical, symbolic, and commonsense reasoning. The dataset includes varying levels of noise in rationales to systematically assess LLMs' performance.
COMPOSITION	
What do the instances that comprise the dataset represent?	Each instance in the NoRa dataset consists of a question, its correct answer, and in-context demonstrations containing noisy reasoning steps.
How many instances are there in total?	The NoRa dataset contains 26,391 questions in total (see Section 3.2). Table 2 contains the sizes of our datasets.
Does the dataset contain all possible instances or is it a sample (not necessarily random) of instances from a larger set?	The NoRa dataset is a carefully curated sample selected from a larger pool of potential reasoning tasks.
What data does each instance consist of?	Each instance in the NoRa dataset contains a question, its correct answer, and in-context demonstrations with noisy reasoning steps.
Is there a label or target associated with each instance?	Yes, each instance has a correct answer that serves as its label.
Is any information missing from individual instances?	No, each instance contains complete information.
Are relationships between individual instances made explicit?	Yes, instances are grouped by reasoning type: mathematical, symbolic, and commonsense.

Are there recommended data splits?	No, this dataset is designed for training-free in-context learning. We provide recommended in-context demonstrations for each question.
Are there any errors, sources of noise, or redundancies in the dataset?	No, the dataset itself is clean and does not contain any errors, unintended noise, or redundancies.
Is the dataset self-contained, or does it link to or otherwise rely on external resources?	Yes, the dataset is self-contained and does not rely on any external resources.
Does the dataset contain data that might be considered confidential?	No.
Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety?	No, the dataset does not include any offensive, insulting, threatening, or anxiety-inducing content.
COLLECTION	
How was the data associated with each instance acquired?	The data was created based on the reference of three rational datasets. (see sec 3.2)
What mechanisms or procedures were used to collect the data?	We systematically added irrelevant or inaccurate thoughts to clean rationales from existing datasets. The irrelevant or inaccurate thoughts are generated from GPT4.
If the dataset is a sample from a larger set, what was the sampling strategy?	Whenever we use subsets, we randomly sample from the original data.
Who was involved in the data collection process and how were they compensated?	The authors of this paper.
Over what timeframe was the data collected?	NoRa was created in 2023. The original datasets of NoRa (Base Calculation, SCAN, and CLUTRR) were created between 2018-2023. We added noisy rationales to these existing datasets to create NoRa.
Were any ethical review processes conducted?	No, as the dataset does not involve human subjects or sensitive information, no formal ethical review was required. However, we adhered to ethical guidelines for AI research throughout the process
PREPROCESSING	
Was any preprocessing/cleaning/labeling of the data done?	Yes. We preprocess the original datasets by adding noisy rationales (irrelevant or inaccurate thoughts) generated by GPT-4 and ensure that the questions and answers remain unchanged. (detailed in Section 3.1)
Was the “raw” data saved in addition to the preprocessed/cleaned/labeled data?	Yes. We retained the original datasets (Base Calculation, SCAN, and CLUTRR) in their raw form, as well as our modified NoRa dataset with added noisy rationales.
Is the software that was used to preprocess/clean/label the data available?	No.
USES	
Has the dataset been used for any tasks already?	Yes, the NoRa dataset has been used in our paper to benchmark the reasoning robustness of large language models against noisy rationales in chain-of-thought prompting.
Is there a repository that links to any or all papers or systems that use the dataset?	Yes. The repository is publicly available at: https://github.com/tmlr-group/NoisyRationales .
What (other) tasks could the dataset be used for?	The NoRa dataset could be used for: 1. Evaluating and improving LLMs’ robustness to noisy information 2. Developing new denoising techniques for reasoning tasks 3. Understanding the impact of In-context noise on LLMs.
Is there anything about the composition of the dataset or the way it was collected and preprocessed/cleaned/labeled that might impact future uses?	The dataset contains artificially introduced noise, which might not perfectly reflect real-world scenarios of noisy reasoning.
Are there tasks for which the dataset should not be used?	The dataset should not be used for tasks that require completely noise-free data or for evaluating models’ performance on clean, standard reasoning tasks without noise.
DISTRIBUTION	
Will the dataset be distributed to third parties outside of the entity on behalf of which the dataset was created?	Yes, we plan to make the dataset available to the research community.
How will the dataset will be distributed?	The dataset will be made available through a public repository (e.g., GitHub) along with necessary documentation for its use.
When will the dataset be distributed?	We aim to release the dataset upon publication of our paper.
Will the dataset be distributed under a copyright or other intellectual property (IP) license, and/or under applicable terms of use (ToU)?	Not to our knowledge.
Have any third parties imposed IP-based or other restrictions on the data associated with the instances?	Not to our knowledge.